

Nested Virtualization_RDP SSH

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Tags

cw.How-To

cw.RDP-SSH

cw.Reviewed-08-2025

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Summary

This article explains how to create a nested environment in Azure to recover a VM when other recovery options have been exhausted and full console access is required for troubleshooting.

References

Public references

- [Nested Virtualization in Azure](#)

- [Introducing the new Dv3 and Ev3 VM sizes](#)

Internal references

- For a video on creating this environment, see [Hyper-V in Azure VMs - Nested Virtualization](#)

Customer enablement

- Manual: [Troubleshoot a problem Azure VM by using nested virtualization in Azure](#)
- Automatic: [Repair a Windows VM by using the Azure Virtual Machine repair commands](#)

Instructions

Azure support teams do not directly support the nested environment itself. We support the host VM as we do any other VM. If there are issues building or using the nested environment, support for the Hyper-V role is provided by the Windows Devices and Deployment team.

Nested Virtualization using az vm repair and CLI

1. Open the Azure portal.
2. Open Azure Cloud Shell to start a CLI session.

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Azure logo, a search bar, and various icons. The left sidebar contains a list of services, with 'Virtual machines' highlighted. The main content area shows the 'Virtual machines' page for a specific subscription. It includes a table with columns for Name, Type, Status, Resource group, Location, and Subscription. Below the table, a PowerShell terminal window is open, displaying the following text:

```

PowerShell
Requesting a Cloud Shell.Succeeded.
Connecting terminal...

MOTD: Modules installed with 'Install-Module' are persisted across sessions

VERBOSE: Authenticating to Azure ...
VERBOSE: Building your Azure drive ...
PS /home/genesis>

```

3. Run the following commands:

```

az account set --subscription "<SubscriptionID>"
az extension add --name vm-repair
az vm repair create --resource-group "<ResourceGroupName>" --name "<VMName>" --enable-nested --verbose

```

Note:

- Replace <ResourceGroupName> and <VMName> with the details of the broken VM.
- When prompted, enter the login credentials. The rescue VM will be deployed in a separate VNet/subnet. To ensure remote access, choose Yes when asked to assign a public IP address.

4. Wait for the rescue VM to be created. This process automatically attaches a copy of the affected disk as a data disk and creates a new resource group. Note the resource group name and the rescue VM name for later steps.
5. Work on the VM as you would with an on-premises VM. Perform any troubleshooting or mitigation steps as needed.
6. After troubleshooting is complete, swap the fixed OS disk (attached to the rescue VM) back to the original VM by running:

```
az vm repair restore --resource-group "<ResourceGroupName>" --name "<VMName>" --verbose
```



Note: Replace <ResourceGroupName> and <VMName> with the details of the broken VM.

Manually create a Rescue VM and provide an IP to the Hyper-V VM to test RDP connectivity

Step 1 — Snapshot the broken disk

1. In the Azure portal, go to **Disks** and select the broken VM's OS disk.

Home > Compute infrastructure | Virtual machines > broken-vm

Compute infrastructure | Virtual machines

Microsoft

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Virtual machines Get started

Overview

All resources

Infrastructure

Virtual machines 1

Virtual Machine Scale Set (VMSS)

Compute Fleet

> Disks + images

> Capacity + placement

> Related services

> Monitoring+Policy

> Help

You are viewing a new version of Browse experience. Some features may be missing. Click here to access the old experience.

Name ↑

broken-vm 2

webServer

broken-vm | Disks

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Operating system

OS disk

Swap OS disk 4

Disk name	Storage type	
broken-vm_OsDisk_1_2a025l	Standard HDD LRS	1

Data disks

Filter by name

Showing 0 of 0 attached data disks

+ Create and attach a new disk Attach existing disk

LUN ⓘ	Disk name	Storage type
No data disks attached		

2. Click **Create snapshot**.



broken-vm_OsDisk_1_2a025b5058454a5ca6173629bd2a31cf

Disk

<< + Create VM + Create VM image version **+ Create snapshot** Delete Refresh Give feedback

Overview

Essentials

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

> Settings

> Monitoring

> Automation

Resource group [\(move\)](#) : [TESTAZURE-RG](#)

Disk state : Attached

Last ownership update time : 5/31/2025, 11:17:18 AM

Location : [South Central US](#)

Subscription [\(move\)](#) : [MCAPS-Support-REQ-74064-2024-riaguila](#)

Subscription ID : 0bc45583-c594-43c5-be5d-6a2285841972

Time created : 5/31/2025, 11:17:18 AM

Tags [\(edit\)](#) : [Add tags](#)

Disk size : 127 GiB

Storage type : Standard HDD LRS

Managed by : [broken-vm](#)

Operating system : Windows

Max shares : 0

Availability zone : No infrastructure redundancy required

Security type : Standard

3. Set the name to Snapshot-BrokenDisk .

4. Set the snapshot type to **Full**, choose **Standard HDD** or **Premium** for storage, and use the same resource group as the rescue VM.

Create snapshot ...

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

MCAPS-Support-REQ-74064-2024-riaguila

Resource group * ⓘ

TESTAZURE-RG

Create new

Instance details

Name *

Snapshot-BrokenDisk ✓

Region ⓘ

(US) South Central US

Snapshot type * ⓘ

☐ **Incremental:** Save on storage costs by making a partial copy of the disk based on the difference between the last snapshot.

☒ **Full:** Make a complete read-only copy of the selected disk.

Source type ⓘ

Disk

Source subscription ⓘ

MCAPS-Support-REQ-74064-2024-riaguila

Source disk ⓘ

broken-vm_OsDisk_1_2a025b5058454a5ca6173629bd2a31cf

Security type ⓘ

Standard

VM generation ⓘ

☐ Generation 1

☒ Generation 2

VM architecture ⓘ

☒ x64

☐ Arm64

5. Click **Create**.

Storage type *

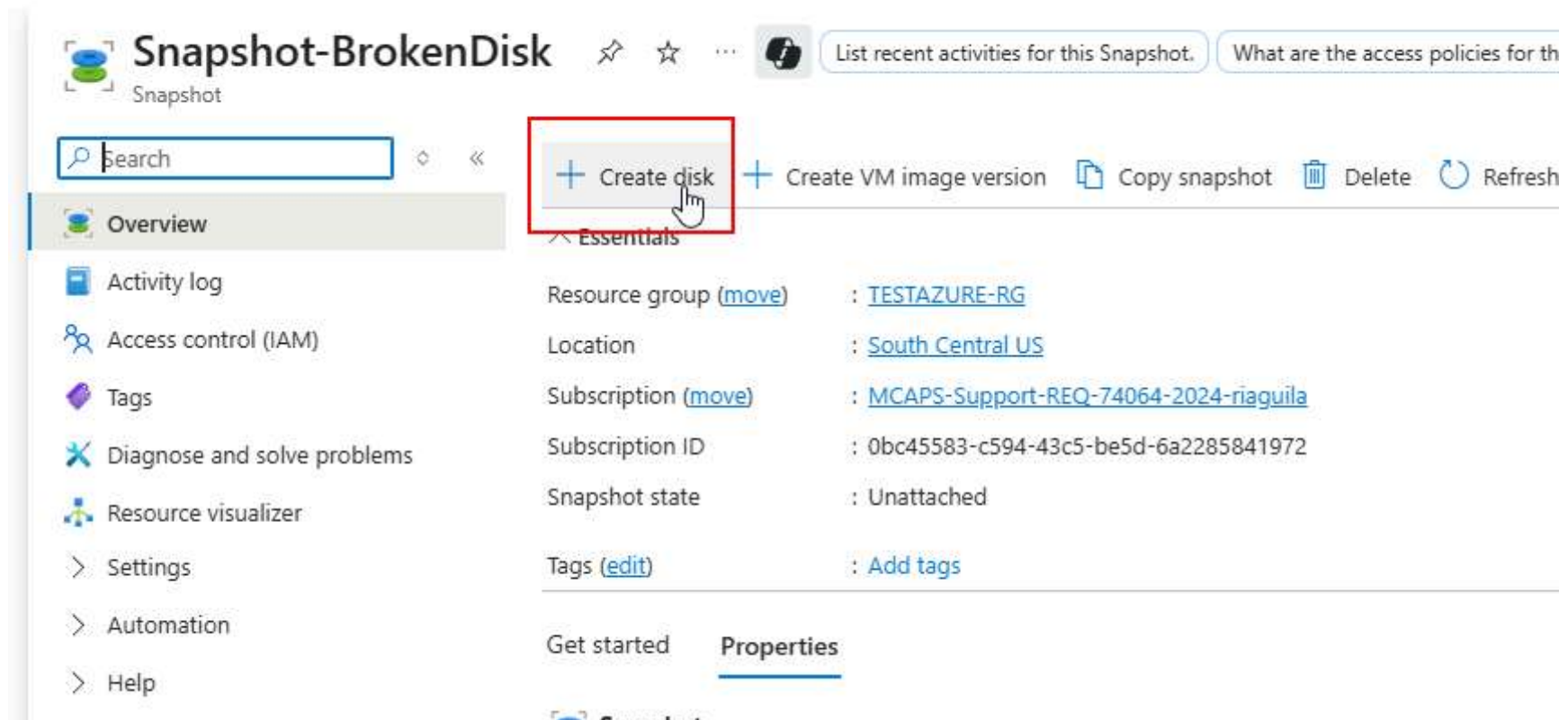
Standard HDD (zone-redundant storage)

Step

1. Go to **Snapshots** and select Snapshot-BrokenDisk.



2. Click **Create disk**.



3. Name the disk `RecoveredDisk` and ensure it is in the same region and resource group as the rescue VM.

Create a managed disk

- Basics
- Encryption
- Networking
- Advanced
- Tags
- Review + create

Select the disk type and size needed for your workload. Azure disks are designed for 99.999% availability. Azure managed disks encrypt your data at rest, by default, using Storage Service Encryption. [Learn more about disks.](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription ⓘ

MCAPS-Support-REQ-74064-2024-riaguila

Resource group ⓘ

TESTAZURE-RG

Create new

Disk details

Disk name ⓘ

RecoveredDisk

Region ⓘ

(US) South Central US

Availability zone

No infrastructure redundancy required

Source type ⓘ

Snapshot

Source subscription ⓘ

MCAPS-Support-REQ-74064-2024-riaguila

Source snapshot ⓘ

Snapshot-BrokenDisk

OS type ⓘ

☐ None (data disk)

☐ Linux

☒ Windows

Security type ⓘ

Standard

VM generation ⓘ

☐ Generation 1

☒ Generation 2

VM architecture ⓘ

☒ x64

☐ Arm64

Size ⓘ

127 GiB

Standard HDD LRS

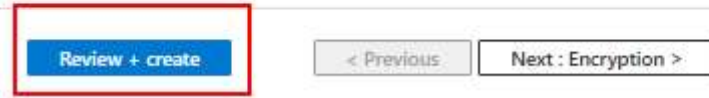
[Change size](#)

4. Click **Create**.



Step

1. In the **Basics** tab, provide the following values:



- **Image:** Windows Server 2019 or 2022
- **Security type:** Standard (disable Trusted Launch)
- **Size:** Select a VM size that supports nested virtualization (for example, Standard_D4s_v3)

Create a virtual machine ...

 Changing Basic options may reset selections you have made. Review all options prior to creating the virtual machine.

-  Help me create a low cost VM
- Help me create a VM optimized for high availability
- Help me choose the right VM size for my workload

Basics Disks Networking Management Monitoring Advanced Tags Review + create

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

MCAPS-Support-REQ-74064-2024-riaguila

Resource group * ⓘ

TESTAZURE-RG

[Create new](#)

Instance details

Virtual machine name * ⓘ

RescueVM

Region * ⓘ

(US) South Central US

Availability options ⓘ

No infrastructure redundancy required

Security type ⓘ

Standard

Image * ⓘ

 Windows Server 2022 Datacenter - x64 Gen2

[See all images](#) | [Configure VM generation](#)

 This image is compatible with additional security features. [Click here to swap to the Trusted launch security type.](#)

VM architecture ⓘ

☐ Arm64

☒ x64

 Arm64 is not supported with the selected image.

☐

Note: Set **Security type** to **Standard**. The default, **Trusted Launch**, does not support nested virtualization. If Trusted Launch is enabled, you will receive an error when adding server roles:

Run with Azure Spot discount (1)

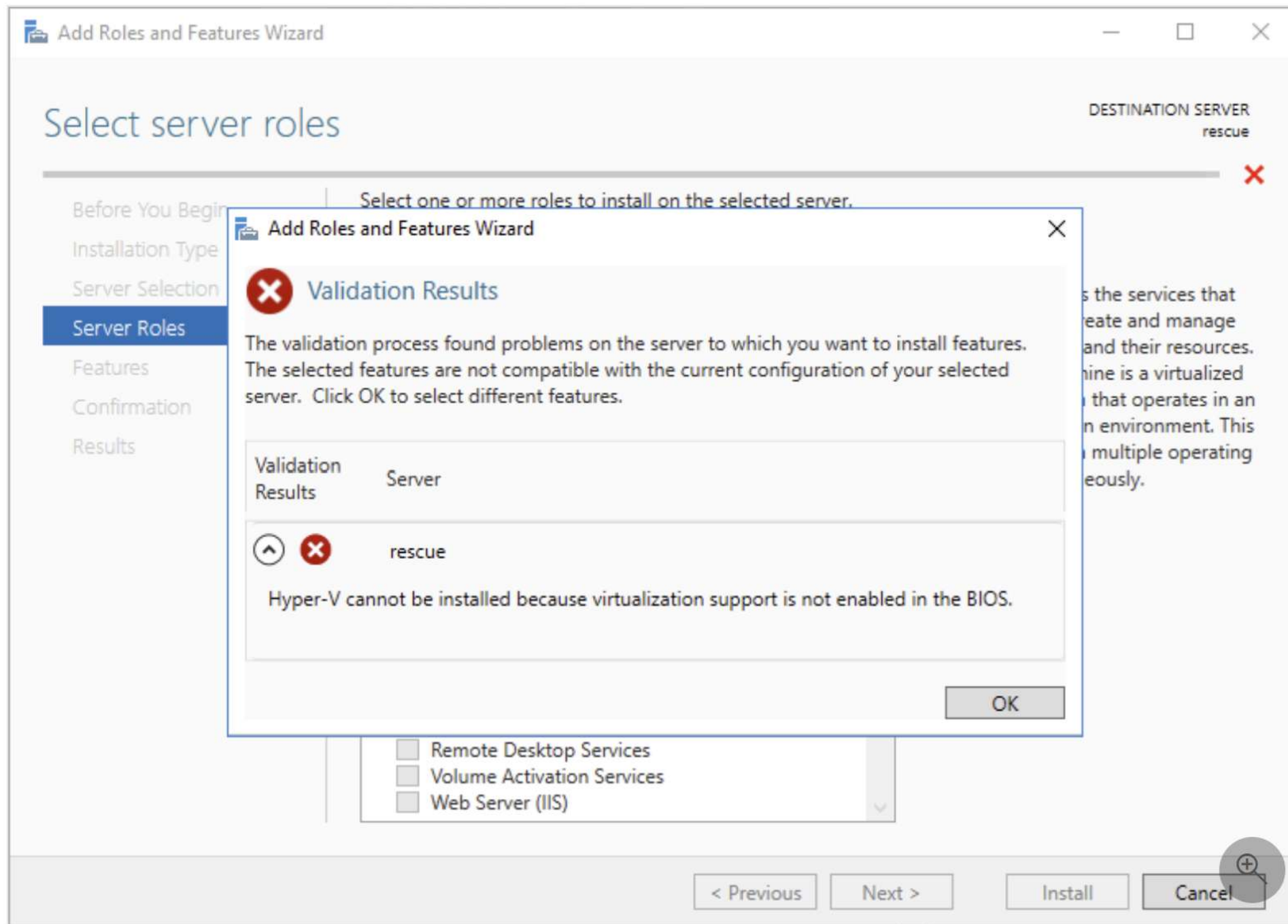
Size

Standard_D2als_v6 - 2 vcpus, 4 GiB memory (\$137.24/month)

[See all sizes](#)



The size you've selected is supported by higher storage performance with NVMe enabled. [Learn more](#)



2. In the **Disks** tab, under **Data disk for RescueVM**, select **Attach existing disk**.

Create a virtual machine ...



Help me create a low cost VM

Help me create a VM optimized for high availability

Help me choose the right VM size for my workload

Basics

Disks

Networking

Management

Monitoring

Advanced

Tags

Review + create

Azure VMs have one operating system disk and a temporary disk for short-term storage. You can attach additional data disks. The size of the VM determines the type of storage you can use and the number of data disks allowed. [Learn more](#)

VM disk encryption

Azure disk storage encryption automatically encrypts your data stored on Azure managed disks (OS and data disks) at rest by default when persisting it to the cloud.

Encryption at host

☐

Encryption at host is not registered for the selected subscription. [Learn more](#)

OS disk

OS disk size

Image default (127 GiB)

OS disk type *

Standard HDD (locally-redundant storage)

The selected VM size supports premium disks. We recommend Premium SSD for high IOPS workloads. Virtual machines with Premium SSD disks qualify for the 99.9% connectivity SLA.

Delete with VM

☒

Key management

Platform-managed key

Enable Ultra Disk compatibility

☐

Data disks for RescueVM

You can add and configure additional data disks for your virtual machine or attach existing disks. This VM also comes with a temporary disk.

LUN	Name	Size (GiB)	Disk type	Host caching	Delete with VM
Create and attach a new disk		Attach an existing disk			



Advanced

Select the disk you created from the broken OS disk:

Azure VMs have one operating system disk and a temporary disk for short-term storage. You can attach additional data disks. The size of the VM determines the type of storage you can use and the number of data disks allowed. [Learn more](#)

VM disk encryption

Azure disk storage encryption automatically encrypts your data stored on Azure managed disks (OS and data disks) at rest by default when persisting it to the cloud.

Encryption at host

☐

Encryption at host is not registered for the selected subscription. [Learn more](#)

OS disk

OS disk size

Image default (127 GiB)

OS disk type *

Standard HDD (locally-redundant storage)

The selected VM size supports premium disks. We recommend Premium SSD for high IOPS workloads. Virtual machines with Premium SSD disks qualify for the 99.9% connectivity SLA.

Delete with VM

☒

Key management

Platform-managed key

Enable Ultra Disk compatibility

☐

Data disks for RescueVM

You can add and configure additional data disks for your virtual machine or attach existing disks. This VM also comes with a temporary disk.

LUN	Name	Size (GiB)	Disk type	Host caching	Delete with VM
0	Select an existin...			None	☐
Create and Advanced	webServer-DiskCopy-20250302204201	size: 127 account type: Standard_LRS disk shares used: 0 of 1			
	webServer-DiskCopy-20250302235934	size: 127 account type: Standard_LRS disk shares used: 0 of 1			
	broken-vm-DiskCopy-20250531174243	size: 127 account type: Standard_LRS disk shares used: 0 of 1			
	RecoveredDisk				

3. Select the appropriate virtual network and accept the default values for other settings unless changes are required.

4. Click **Create**.



Step 4: Access the Disk

1. Connect to RescueVM via RDP.

2. Open **Disk Management**, bring the disk online, and assign a drive letter if one is not assigned automatically.

Disk Management

File Action View Help

Volume	Layout	Type	File System	Status	Capacity	Free Sp...	% Free
(Disk 0 partition 3)	Simple	Basic		Healthy (E...	99 MB	99 MB	100 %
(Disk 1 partition 3)	Simple	Basic		Healthy (E...	99 MB	99 MB	100 %
Windows (C:)	Simple	Basic	NTFS	Healthy (B...	126.45 GB	114.60 GB	91 %

Disk 0

Basic

126.99 GB

Online

450 MB

99 MB

Healthy (EFI System I

Windows (C:)

126.45 GB NTFS

Healthy (Boot, Page File, Crash Dump, Basic Data Partition,

Disk 1

Basic

126.99 GB

Offline

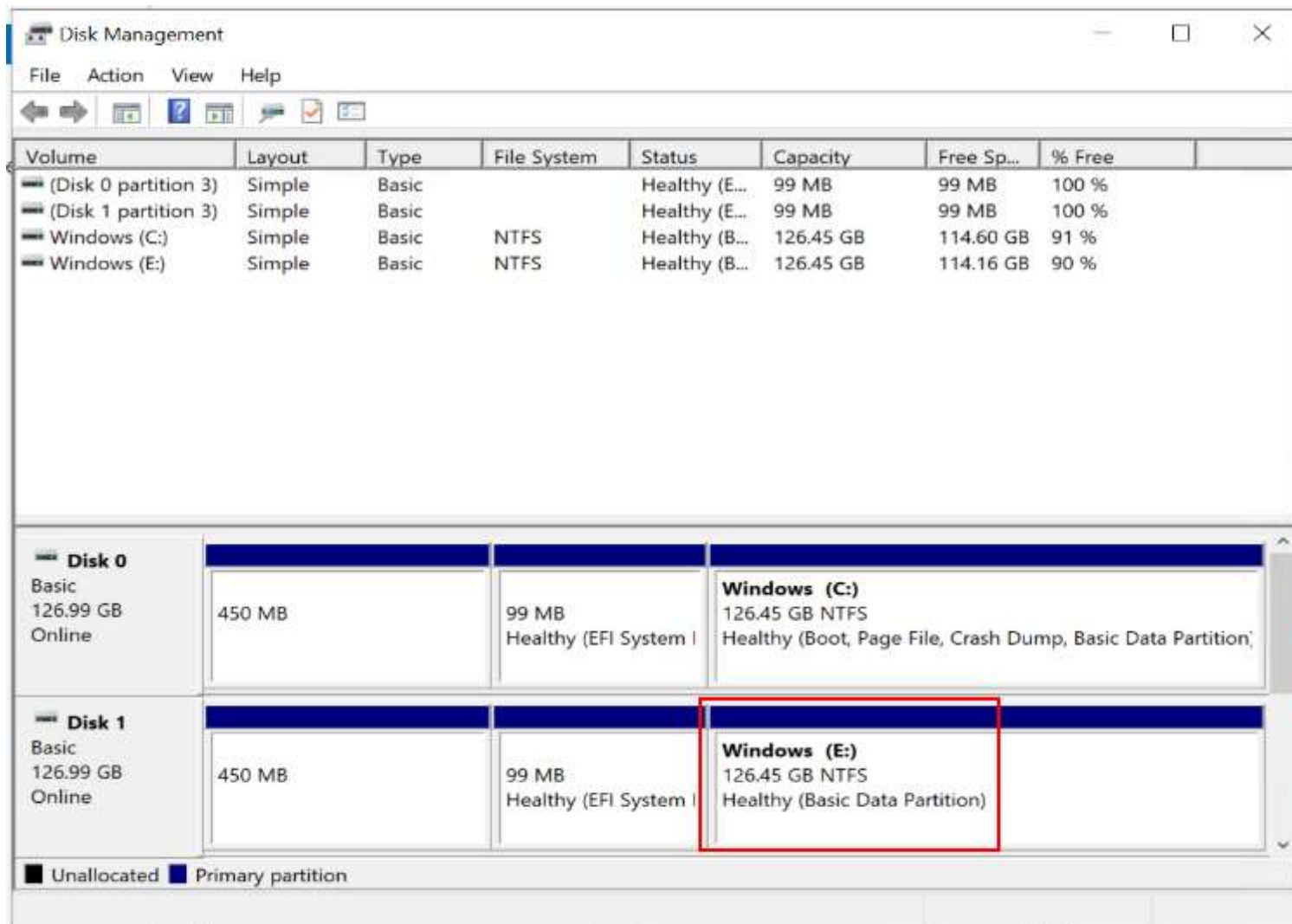
Online

Properties

Help

Unallocated

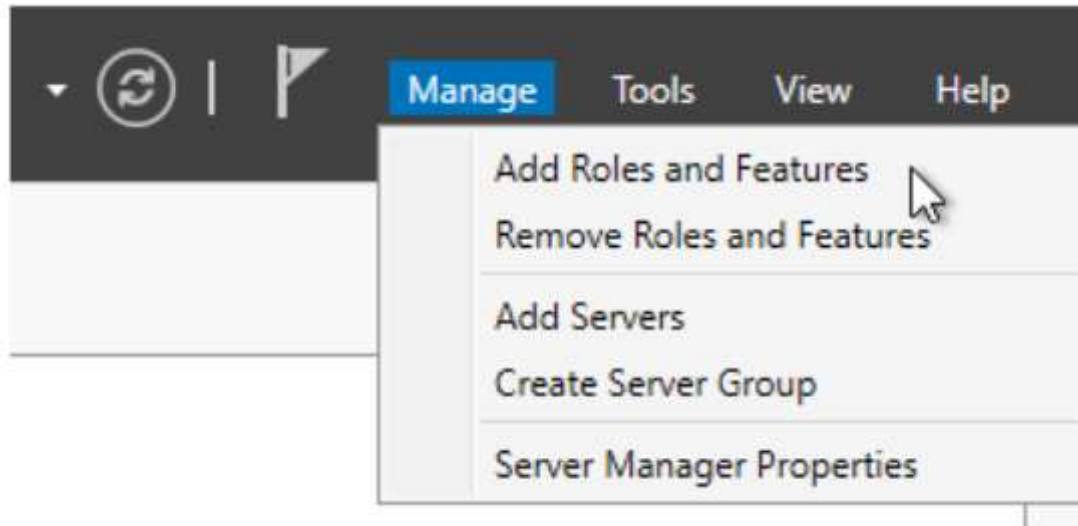
Primary partition



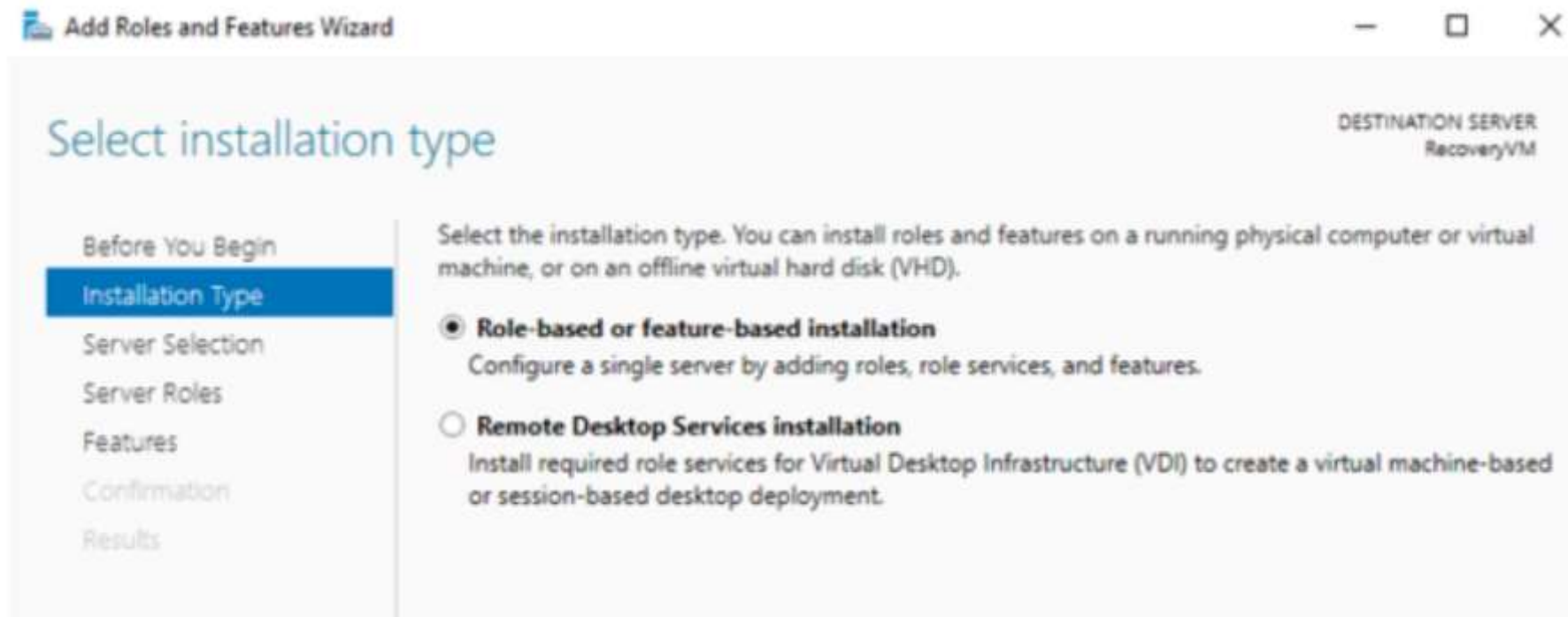
3. Recover or troubleshoot data as needed. If you require guest-OS access via nested virtualization, continue to Step 5.

Step 5: Install Hyper-V on the Rescue VM

1. In **Server Manager**, select **Manage > Add Roles and Features**.



2. For **Installation Type**, choose **Role-based or feature-based installation**.



3. Ensure RescueVM is selected.

Select destination server

DESTINATION SERVER
RecoveryVM

Before You Begin

Installation Type

Server Selection

Server Roles

Features

Confirmation

Results

Select a server or a virtual hard disk on which to install roles and features.

☒ Select a server from the server pool☐ Select a virtual hard disk

Server Pool

Filter:

Name	IP Address	Operating System
RecoveryVM	10.1.2.5	Microsoft Windows Server 2016 Datacenter

4. Select the **Hyper-V** role.

Select server roles

Before You Begin

Installation Type

Server Selection

Server Roles

Features

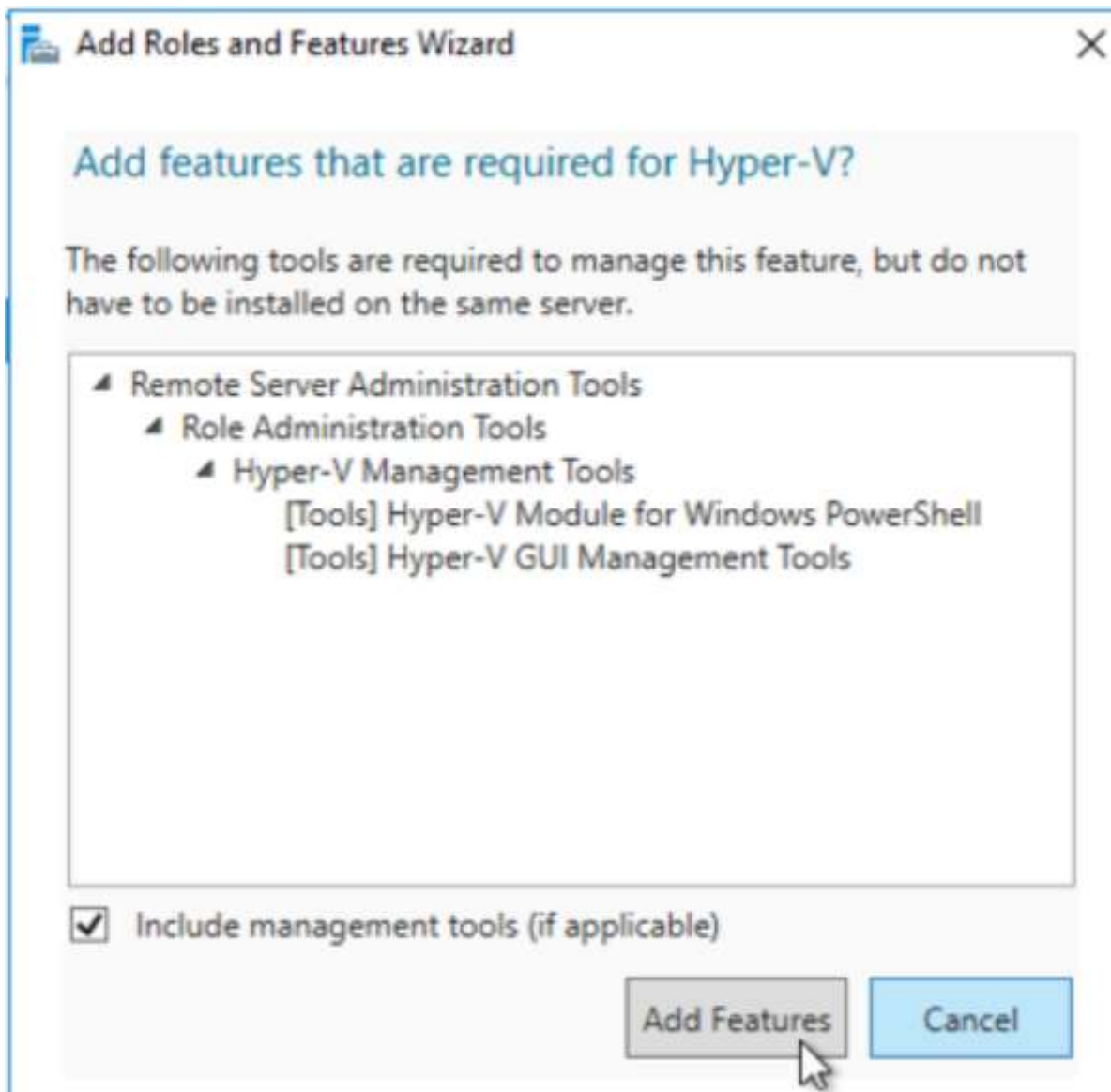
Confirmation

Results

Select one or more roles to install on the selected server.

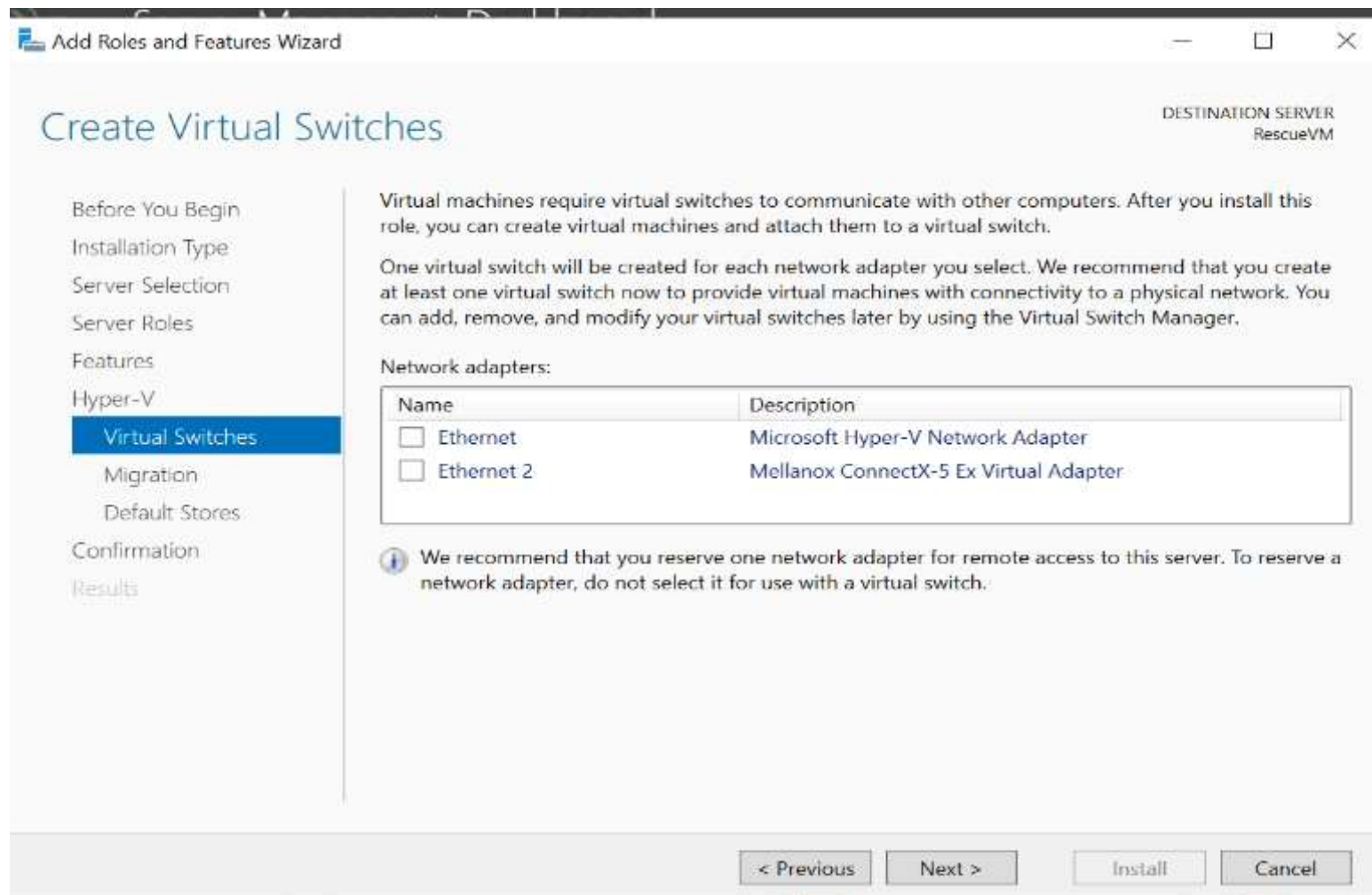
Roles

- ☐ Active Directory Certificate Services
- ☐ Active Directory Domain Services
- ☐ Active Directory Federation Services
- ☐ Active Directory Lightweight Directory Services
- ☐ Active Directory Rights Management Services
- ☐ Device Health Attestation
- ☐ DHCP Server
- ☐ DNS Server
- ☐ Fax Server
- ☒ File and Storage Services (1 of 12 installed)
- ☐ Host Guardian Service
- ☐ Hyper-V
- ☐ MultiPoint Services
- ☐ Network Controller
- ☐ Network Policy and Access Services
- ☐ Print and Document Services
- ☐ Remote Access
- ☐ Remote Desktop Services
- ☐ Volume Activation Services
- ☐ Web Server (IIS)



5. Click **Next** on the Features screen.

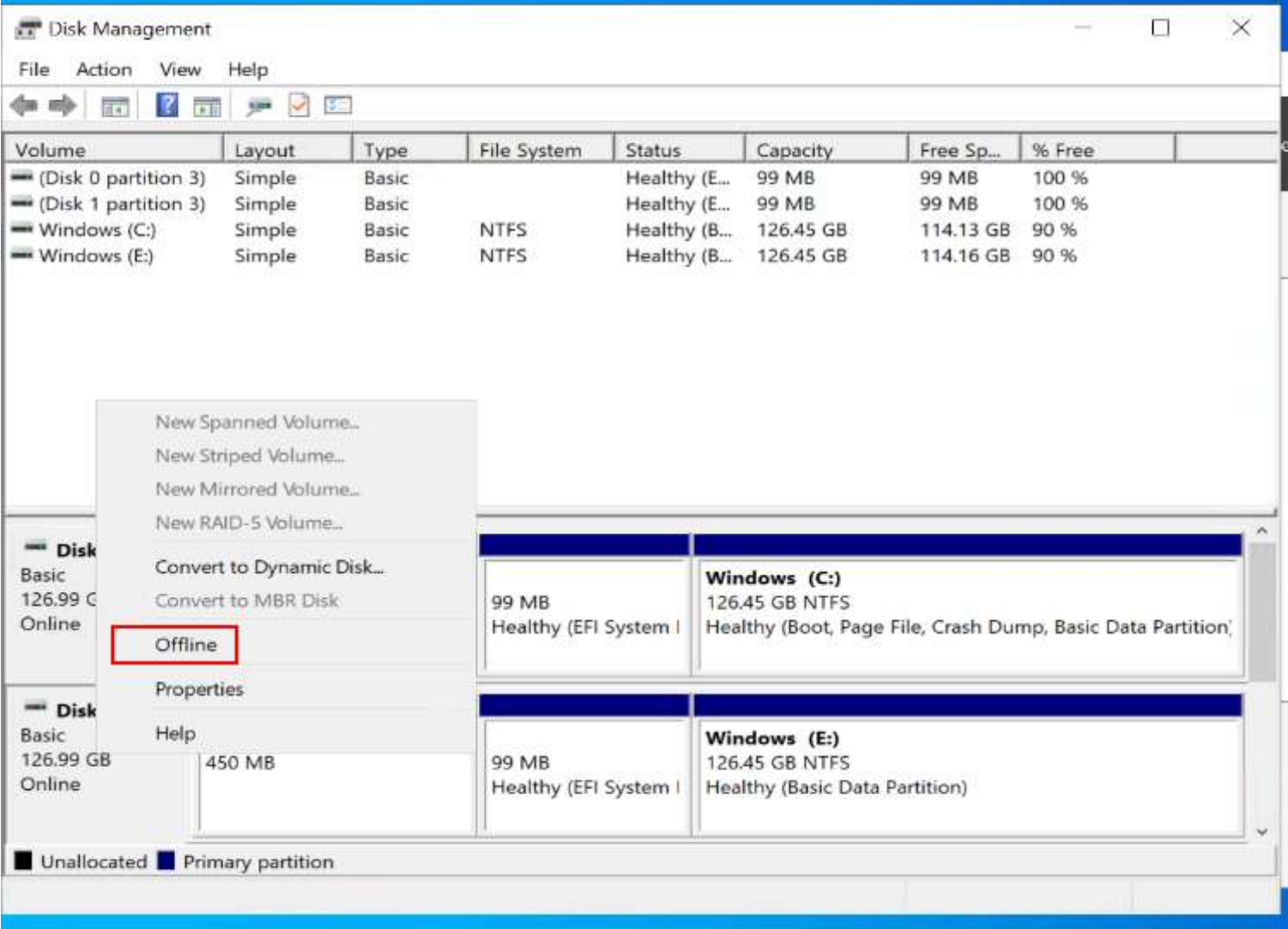
6. Click **Next** on the Virtual Switches screen.



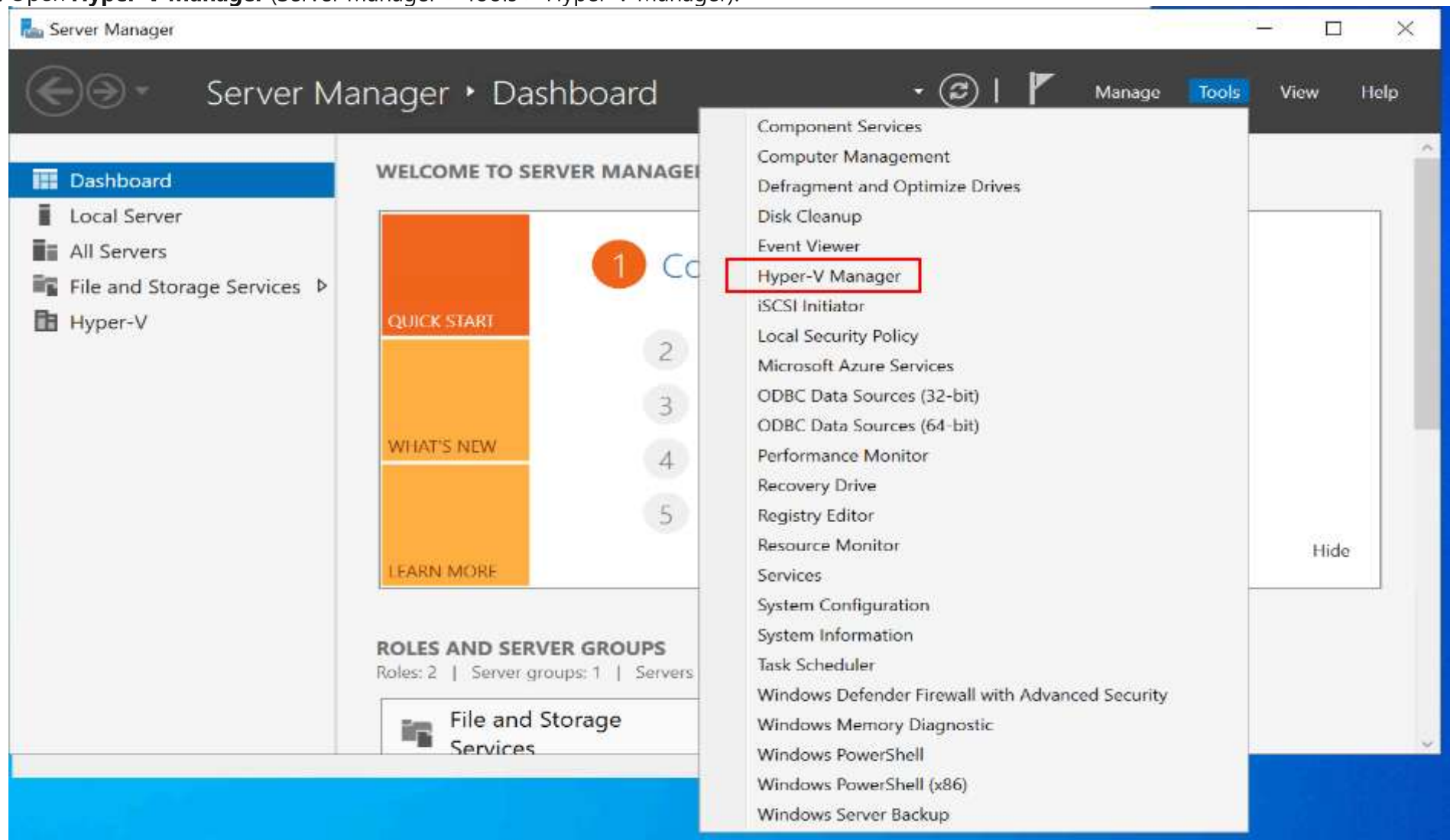
7. Click **Next** on the Migration page.
8. Click **Next** on the Default Stores page.
9. Select the option to restart the server automatically if required.
10. Click **Install**.
11. Allow the server to install the Hyper-V role. This may take several minutes; the server will reboot automatically. Monitor the reboot using the **Boot Diagnostics** tab in the Azure portal.

Step 6 — Create the Hyper-V VM in Hyper-V Manager

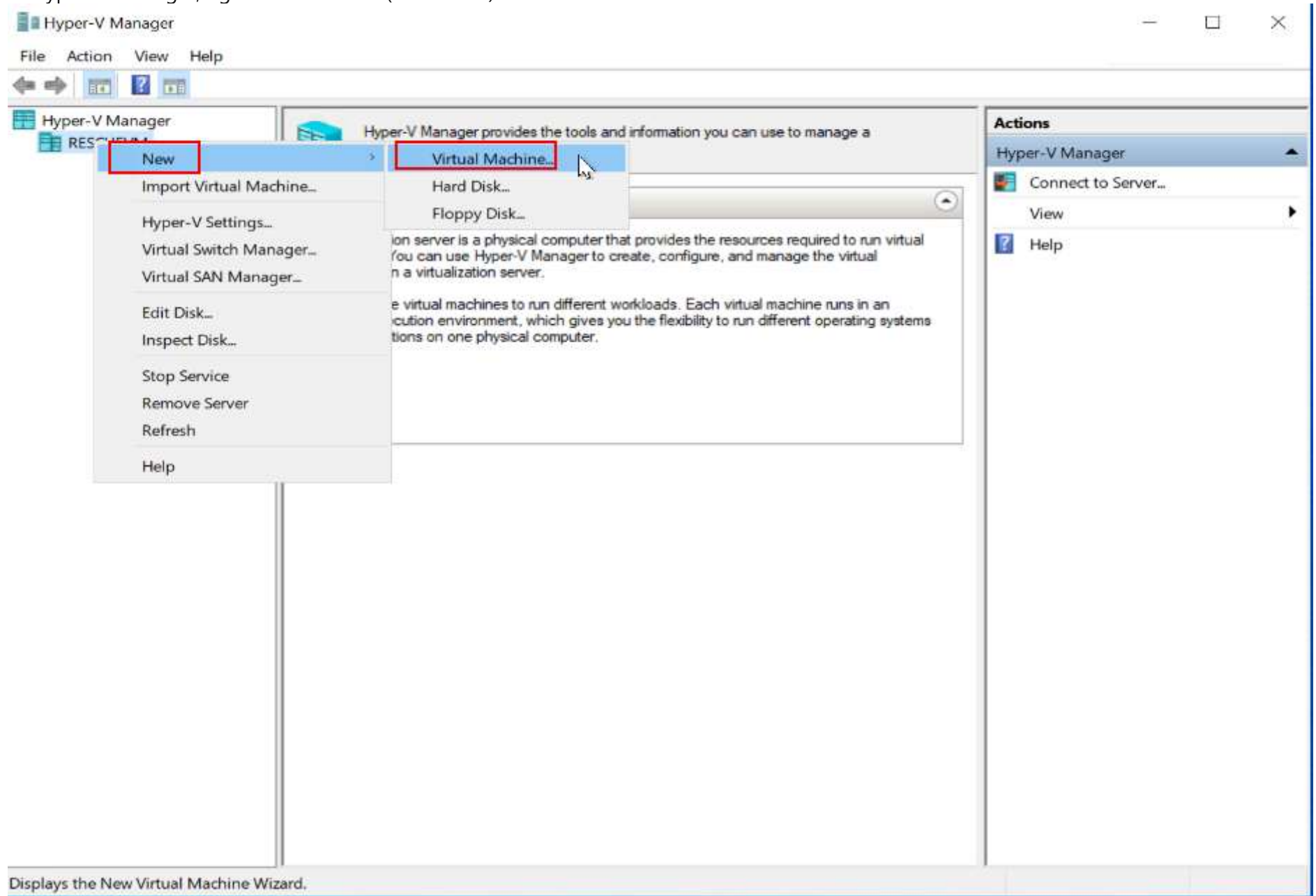
1. Open **Disk Management** and ensure the VHD of the broken VM is set to Offline.



2. Open **Hyper-V Manager** (Server Manager > Tools > Hyper-V Manager).

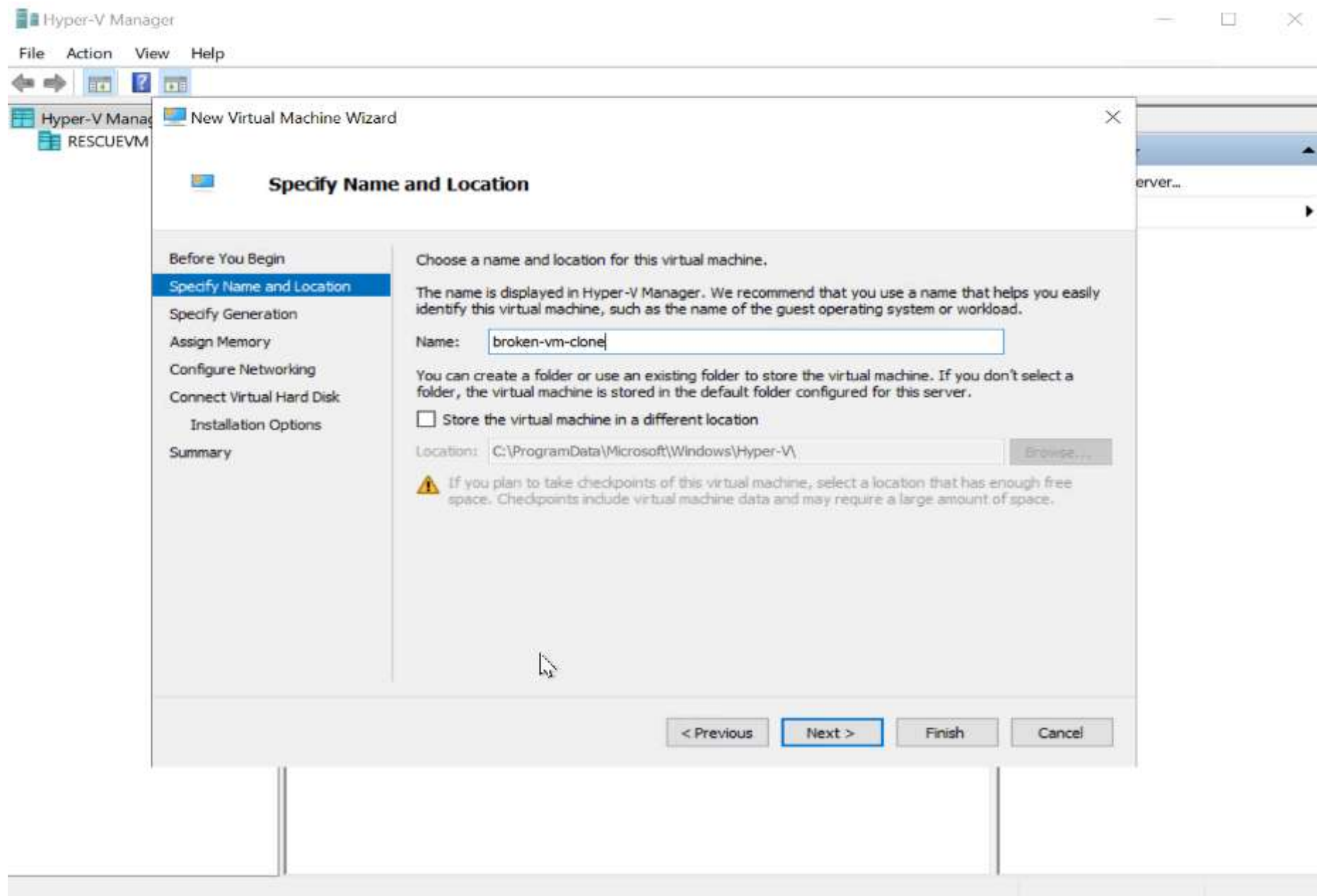


3. In Hyper-V Manager, right-click the host (RescueVM) and choose **New > Virtual Machine**.



4. In the New Virtual Machine wizard, click **Next**.

5. Enter a name for the VM and click **Next**.

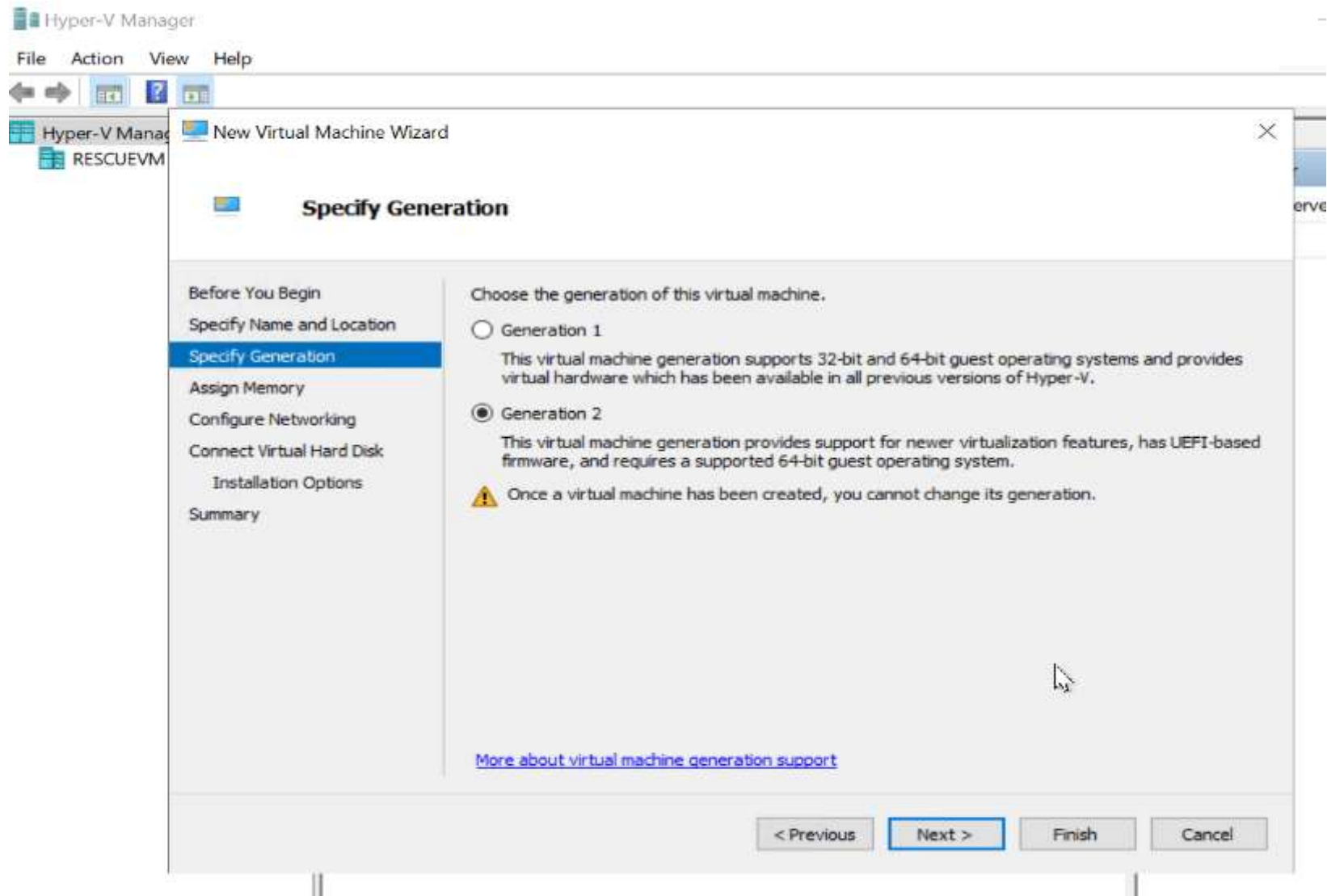


6. Select the generation that matches the source VM (for example, **Generation 2**).



Virtual machine

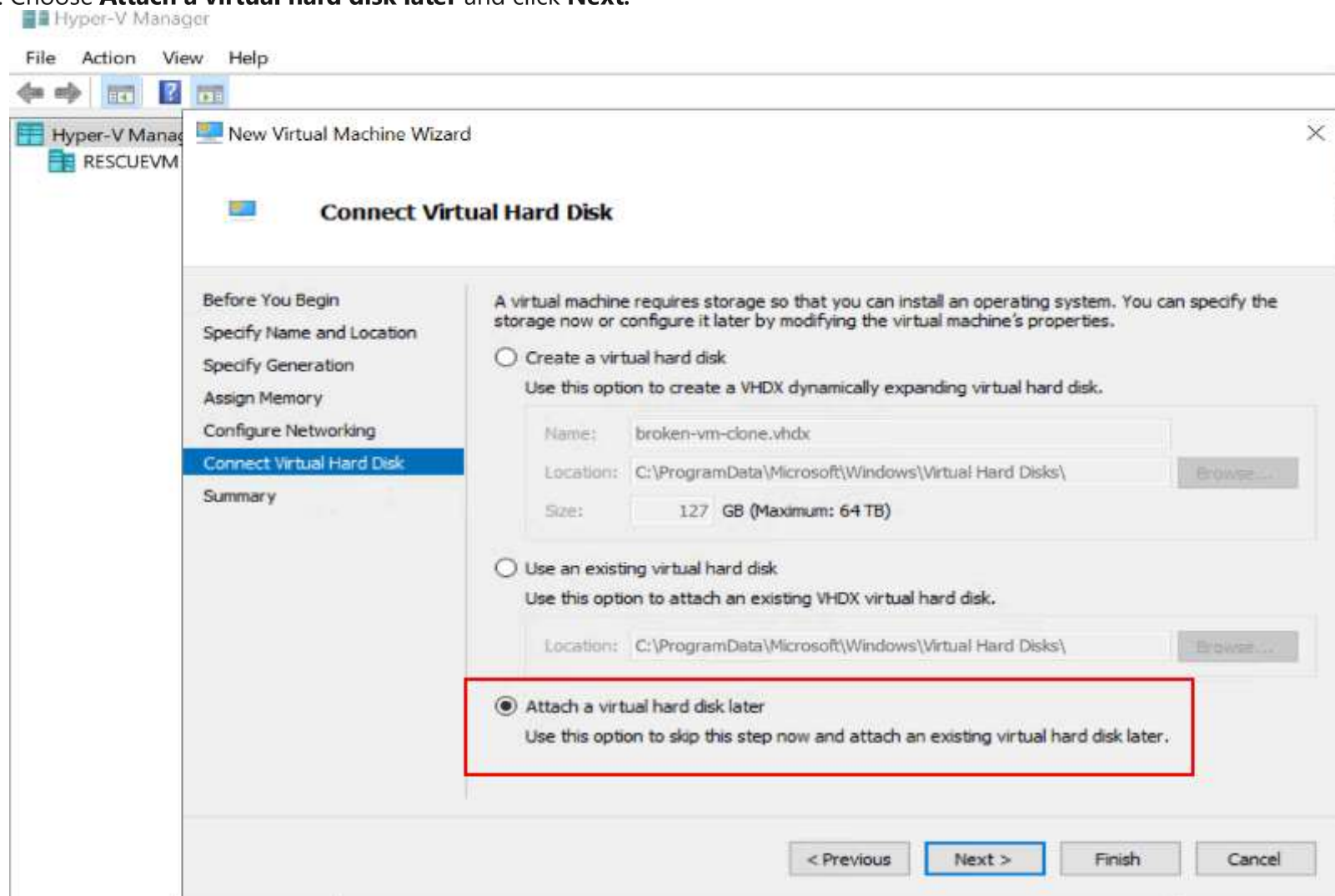
Computer name	broken-vm
Operating system	Windows (Windows Server 2022 Datacenter)
VM generation	V2



7. Set the startup memory to **4096 MB** (or as required) and click **Next**.

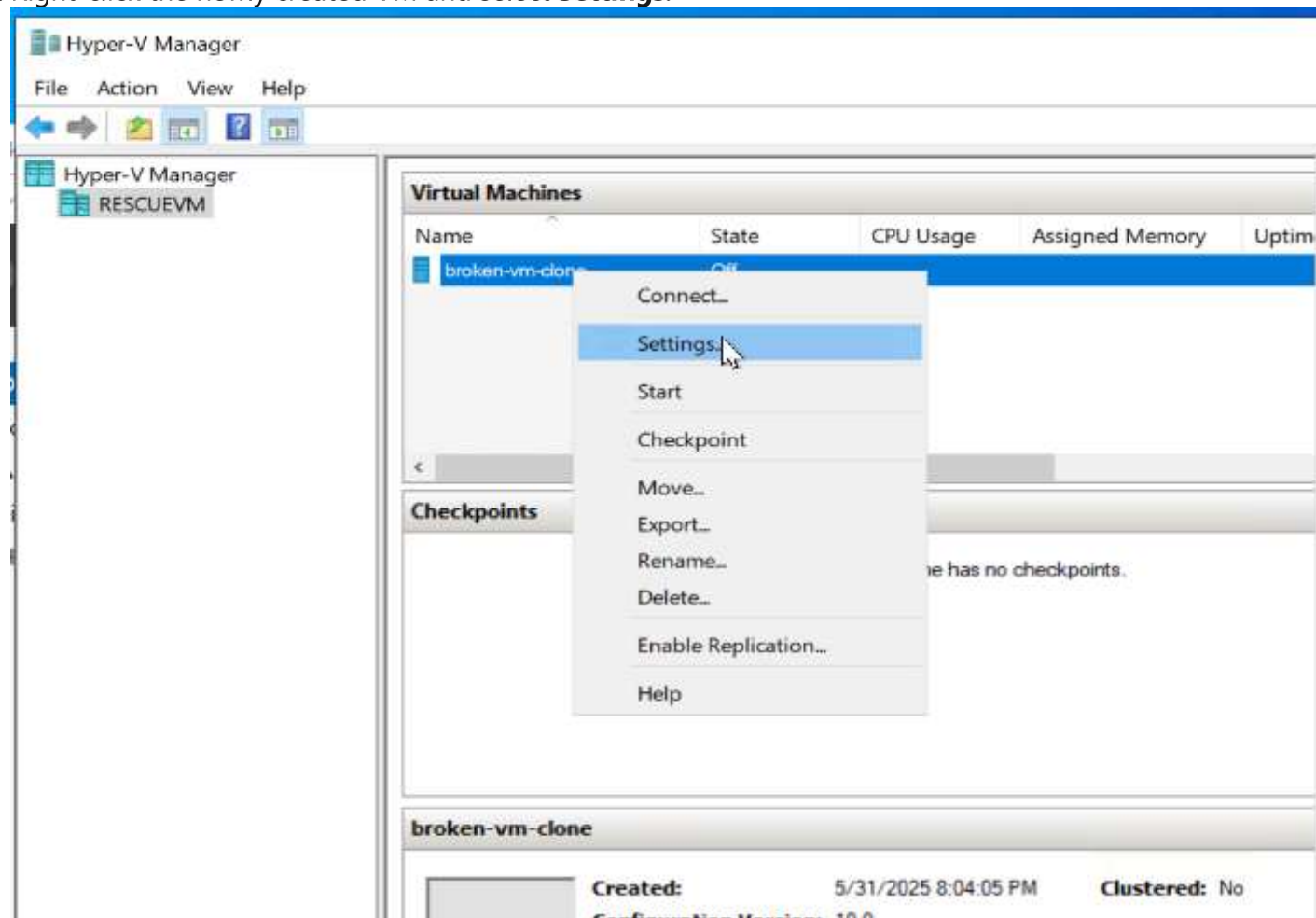
8. On Configure Networking, choose **Not connected**.

9. Choose **Attach a virtual hard disk later** and click **Next**.



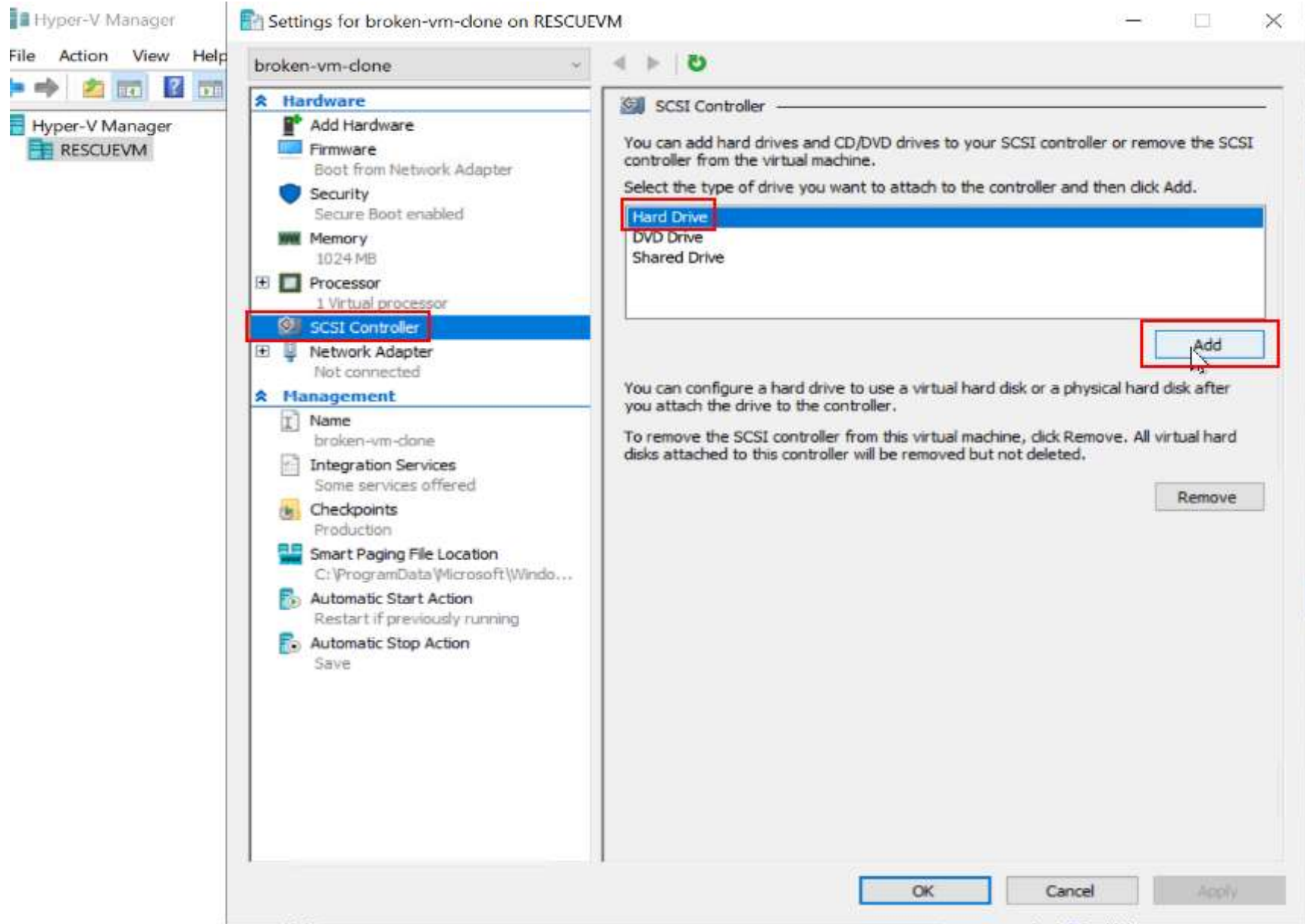
10. Click **Finish** to create the VM.

11. Right-click the newly created VM and select **Settings**.



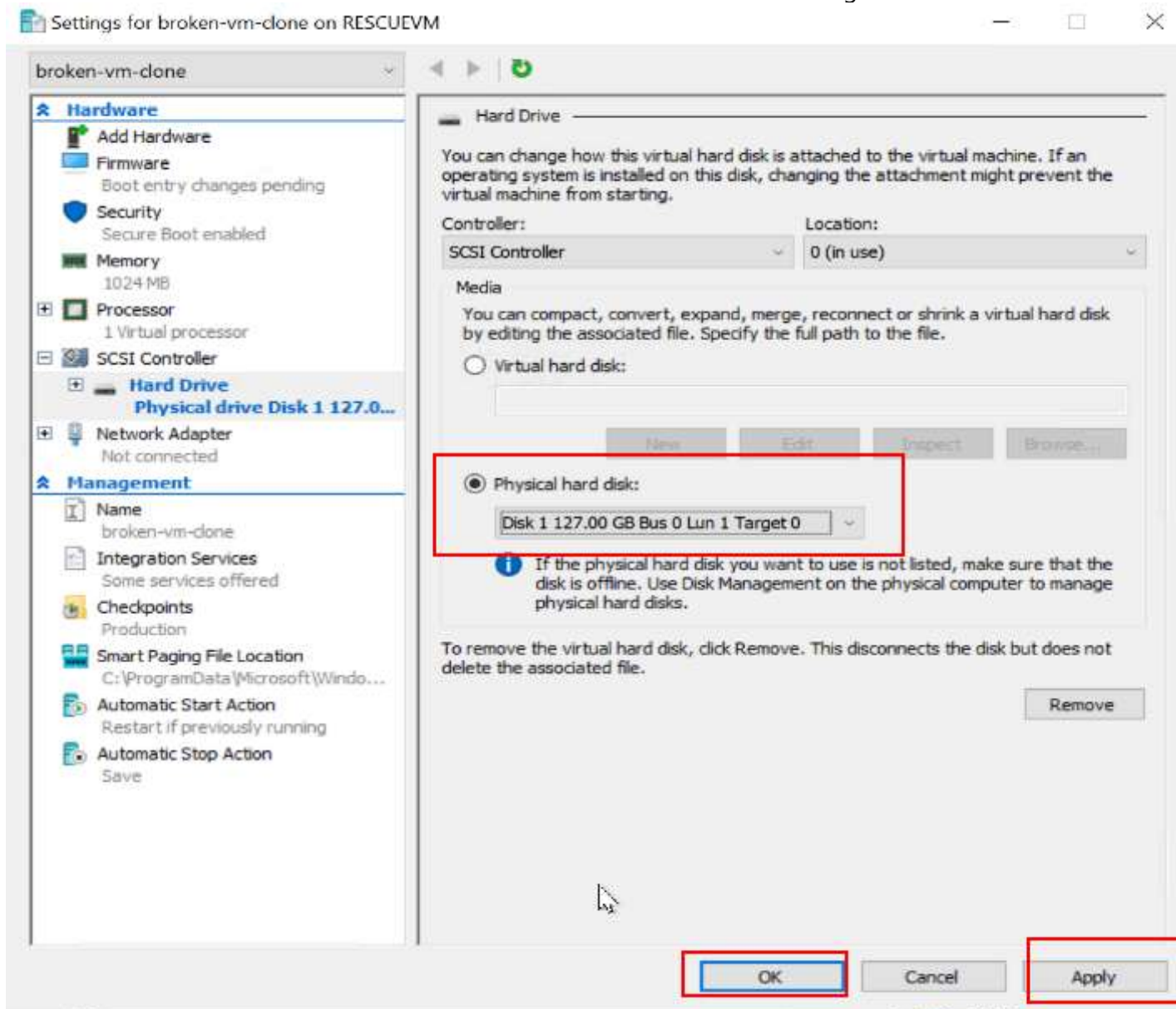
12. For a Generation 2 VM, select **SCSI Controller** (for Gen 1, select **IDE Controller**).

13. Select **Hard Drive** and click **Add**.



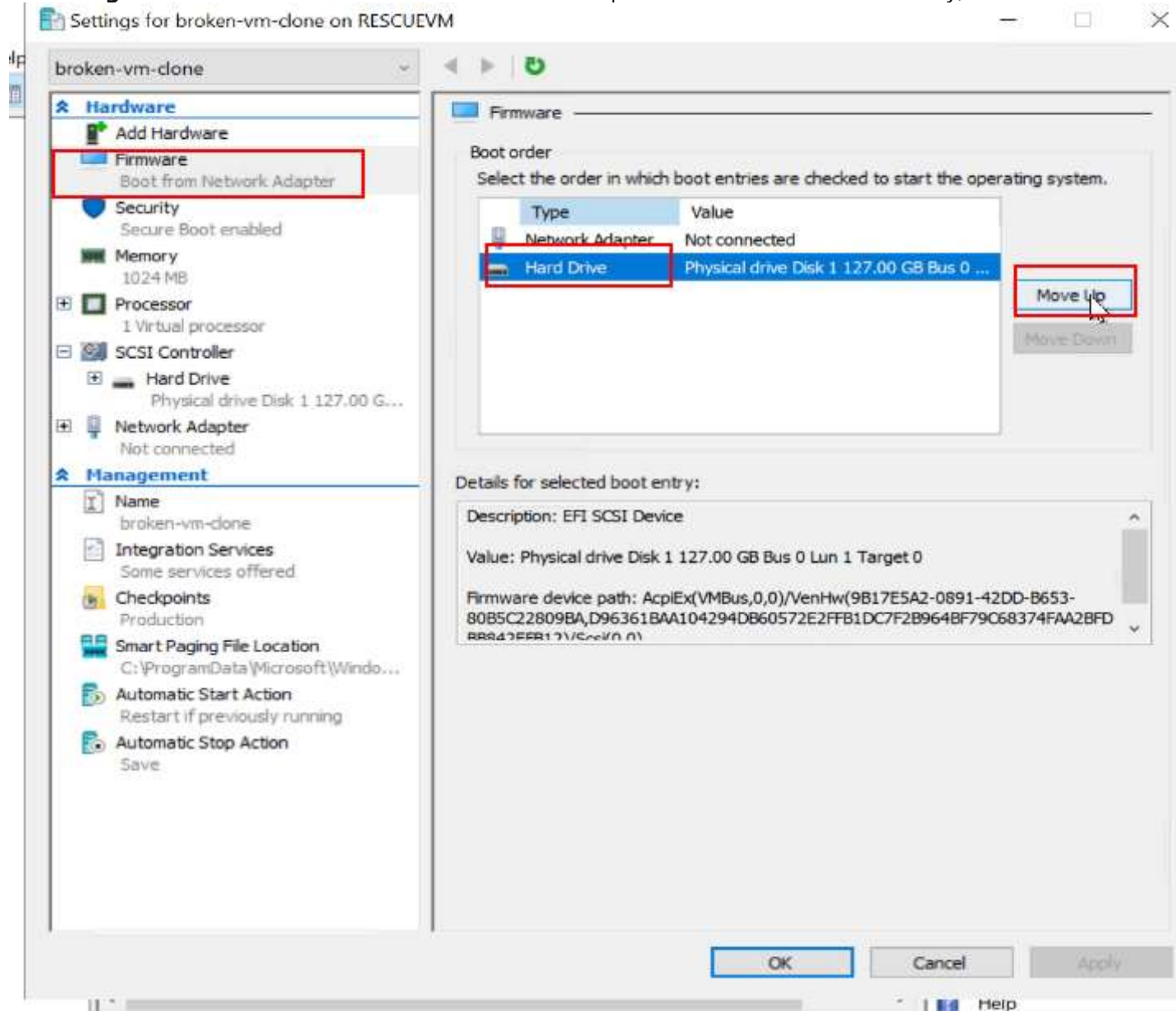
14. Under **Physical Hard Disk**, select the VHD of the broken VM that you attached to the Rescue VM.

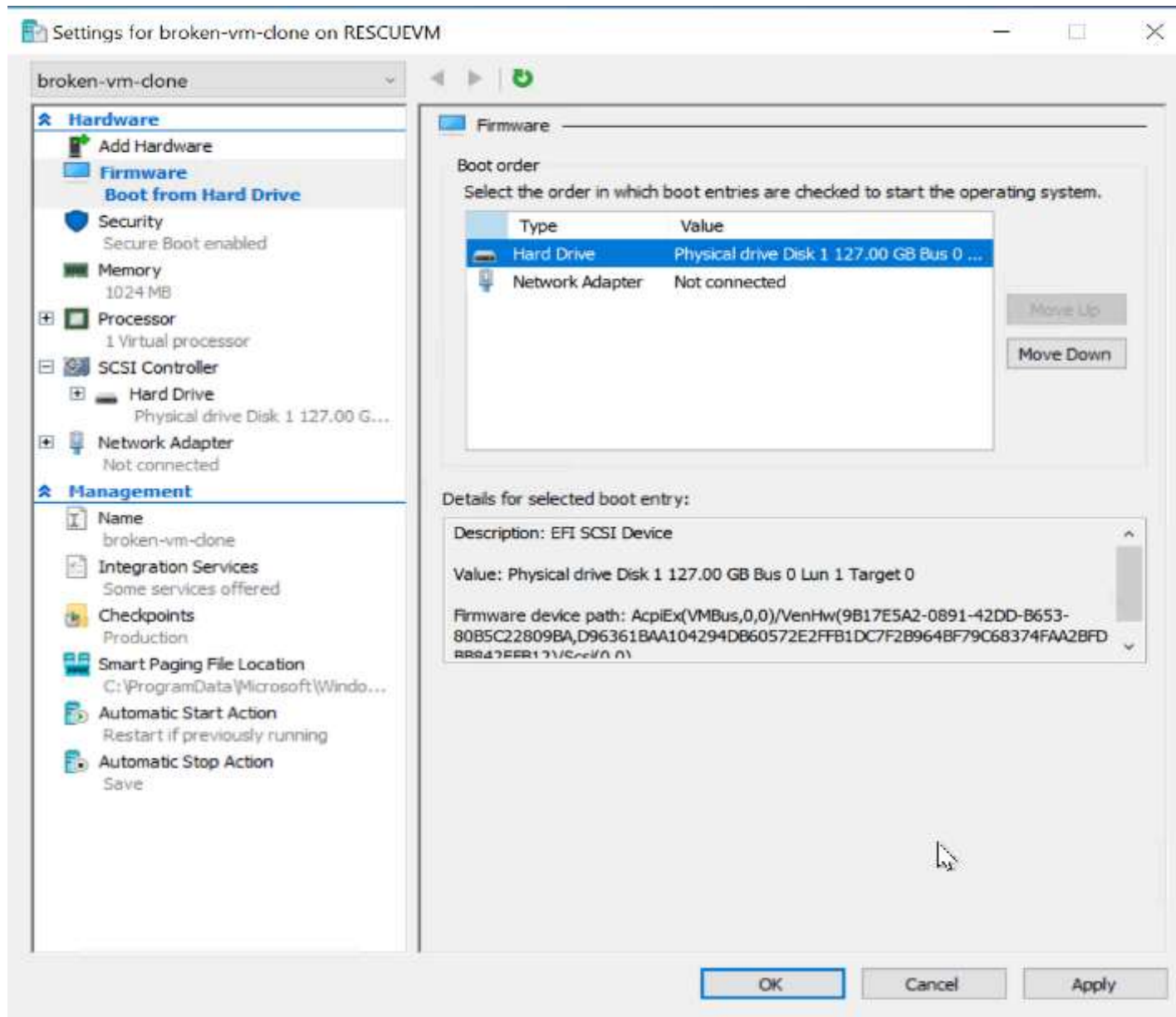
15. If no disks are listed, ensure the attached VHD is set to **Offline** in Disk Management on the Rescue VM.



16. Click **Apply** and **OK**.

17. In **Settings > Firmware**, ensure **Hard Drive** is at the top of the boot order. If necessary, select it and click **Move Up**.





18. Click **Apply** and **OK**, then double-click the VM and click **Start**.

You can now troubleshoot the nested VM as you would an on-premises machine.

Step 7 (Optional): Enable inbound/outbound access to the nested VM

Reference: See the TSG [Enable Inbound/Outbound Access for Nested VM](#) for instructions.

Step 8 — Swap the broken OS disk with the fixed disk

1. After verifying the VM is fixed, shut down the nested VM in Hyper-V Manager.
2. In the Azure portal, open the Rescue VM and go to **Disks**.
3. Detach the fixed VHD from the Rescue VM.

[Home](#) > [CreateDiskBlade-20250531142214](#) | [Overview](#) > [TESTAZURE-RG](#) > [RescueVM](#)

RescueVM | Disks

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Swap OS disk

Disk name	Storage type	Size (GiB)	Max IOPS	Max throughput (...)	Encryption ⓘ	Host caching ⓘ
RescueVM_disk1_778af315d47e4dce9e4dbce71f94b638	Standard HDD LRS	127	500	60	SSE with PMK	Read/write

Data disks

Filter by name

Showing 1 of 1 attached data disks

Create and attach a new disk

Attach existing disks

LUN ⓘ	Disk name	Storage type	Size (GiB)	Max IOPS	Max throughput (...)	Encryption ⓘ	Host caching ⓘ
0	websvrdisk-clone	Standard HDD LRS	127	500	60	SSE with PMK	None



RescueVM | Disks

☆ ...

Virtual machine



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 - Extensions + applications

OS disk

 Swap OS disk

Disk name	Storage type
RescueVM_disk1_778af315d47e4dce9e4dbce71f94b638	Standard HDD LRS

Data disks

 Filter by name

Showing 0 of 0 attached data disks

 Create and attach a new disk  Attach existing disks

LUN ⓘ	Disk name	Storage type
No data disks attached		

4. Replace the OS disk on the original (broken) VM: a. Navigate to the broken VM > **Disks** > **Swap OS disk**.

Home > broken-vm

broken-vm | Disks

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OS disk

Swap OS disk

Disk name	Storage type	Size (GiB)
broken-vm_OsDisk_1_2a025b5058454a5ca6173629bd2a31cf	Standard HDD LRS	127

Data disks

Filter by name

Showing 0 of 0 attached data disks

+ Create and attach a new disk Attach existing disks

LUN ⓘ	Disk name	Storage type	Size (GiB)
No data disks attached			

b. Select the fixed disk, type the VM name to confirm, and click **OK**.

Swap OS Disk ...

Swap the OS disk for a backup disk or another disk for VM troubleshooting, [Learn more](#).

Choose disk *

Select an existing disk



All disks

webServer-DiskCopy-20250302204201

size: 127 GiB, account type: Standard HDD LRS, resource group: TEST-CUSTOMFQDN-RG, disk shares ...

webServer-DiskCopy-20250302235934

size: 127 GiB, account type: Standard HDD LRS, resource group: TEST-CUSTOMFQDN-RG, disk shares ...

broken-vm-DiskCopy-20250531174243

size: 127 GiB, account type: Standard HDD LRS, resource group: TESTAZURE-RG, disk shares used 0 of ...

Disks in resource group 'TESTAZURE-RG'

broken-vm-DiskCopy-20250531174243

size: 127 GiB, account type: Standard HDD LRS, disk shares used 0 of 1 

[Home](#) > [broken-vm](#) | [Disks](#) >

Swap OS Disk ...

Swap the OS disk for a backup disk or another disk for VM troubleshooting, [Learn more](#).

Choose disk *

broken-vm-DiskCopy-20250531174243



This VM will be stopped (deallocated) and the OS disk will be replaced. Any existing data on the OS disk will be lost.

Confirm you want to swap the OS disk for this VM by entering the name of the vm 'broken-vm'

broken-vm



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If the customer requests a credit refund for the extra work involved in creating the nested environment, follow this process:



- 1. Open a problem ticket to Billing with the following details in the issue description area:
 - **Product:** Azure Billing
 - **Support topic:** Credit request / Help with credit request for a scenario not listed
 - **Issue description:**

Subscription ID for the affected subscription:
Services affected (SQL Server, Virtual Machine):
Instance ID:
VM Size:
Problem description:
Cause:
Effect:
Issue start (in UTC):
Issue end (in UTC):



Need additional help or have feedback?

To engage the Azure RDP-SSH SMEs...	To provide feedback on this page...	To provide kudos on this page...
If you are in the OneVM team then reach out to the RDP-SSH SMEs AVA channel via Teams. If you are a VM Delivery Partner then follow your team's process for swarming or ask your TA. Make sure to use the Ava process for faster assistance.	Use the RDP-SSH Feedback form to submit detailed feedback on improvements or new content ideas for RDP-SSH. Please note the link to the page is required when submitting feedback on existing pages! If it is a new content idea, please put N/A in the Wiki Page Link.	Use the RDP-SSH Kudos form to submit kudos on the page. Kudos will help us improve our wiki content overall! Please note the link to the page is required when submitting kudos!